Resolution Geometry: Coherence Geometry

Does Coherence Deserve Its Own Geometry? Yes, Conditionally: the Fibre over Uncertainty, with Connection, Curvature and a Selected Metric

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Abstract

Uncertainty geometry is the Fisher geometry of what a protocol resolves. Čencov's theorem forces its metric, and it exists for every realization. We ask whether coherence is merely a region of that geometry, or whether it deserves a geometry of its own. We fix six criteria in advance (S1–S6) and attack each with a seeded battery.

Verdict. Coherence passes all six, under two conditions: the record cone is not a simplex, and a reference frame is declared.

- *Existence and identification.* A maximal reference protocol is complete iff the record cone is a simplex. Otherwise it leaves exactly $k\,d(d-1)/2$ directions unresolved: the Peirce off-diagonal spaces, with $k = 1, 2, 4$ for real, complex and quaternionic cones. These directions are invisible to population records, to partial quadratures and to every scalar coherence value. Two quadrature settings per pair identify them completely.
- *Bundle structure.* Forgetting the reference phase is a quotient (dephasing). It equals the twirl over the frame gauge group $\mathbb{Z}_2$, $U(1)$ or $Sp(1)$. Coherence is the fibre of this quotient over the population simplex, and the simplex carries the uncertainty geometry.
- *Connection and holonomy.* Relative to a frame, coherence is a connection on the support graph. Its gauge classes are exactly: populations, magnitudes, support graph, and holonomy on $b_1 = |E| - |V| + c$ independent cycles. The holonomy is physical: it changes which magnitudes positivity allows, from $\frac{1}{3}$ down to $\frac{1}{6}$.
- *Metric.* Čencov uniqueness holds on the base and fails exactly on the fibre, where the monotone metrics form the Petz family. So the entire task dependence of the quantum metric lives in the coherence fibre. RG's resolution task, the supremum over protocols of record Fisher information, selects the SLD (Bures) metric uniquely. Stein and Chernoff testing select BKM and Wigner–Yanase, as in the print's task lock.
- *Curvature and topology.* The pure stratum is Kähler and toric: populations are the moment map of the frame torus, so fixed-population pure fibres are Lagrangian tori and the symplectic form pairs populations with phases (action–angle). The Berry–Uhlmann curvature, with Chern number $|C| = 1$ on the pure qubit stratum, therefore lives on the total space, not on a fibre. Curvature obstructs joint resolution (Gill–Massar). The Schrödinger flow is the Hamiltonian flow of $\langle H \rangle$.
- *Morphisms.* Relative-entropy coherence is monotone under a channel iff the channel creates no coherence from incoherent states (MIO).

RG reading. Coherence is the resolvability of the reference group's orbit. It is the obstruction to treating a reference relabeling as null.

Scope and evidence.

- Classical partially coherent waves realize the same geometry. Coherence geometry is therefore a property of the record cone, not of quantum ontology.
- The complex structure group is selected only together with a composition rule.
- No new mathematical primitive is introduced.
- Fifteen groups (K1–K15), each with a canary, rerun byte-identically.

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1 The question and the test

Uncertainty Geometry (RG print) takes the indistinguishability of declared protocols as its primitive. Its metric is forced, the Fisher metric by Čencov's theorem, and it exists in every realization. Coherence is usually introduced as a basis-dependent number. We ask a sharper question: is there a *geometry* that belongs to coherence, or is coherence only a set of coordinates on the uncertainty geometry?

To prevent the answer from being chosen after the fact, the test is fixed in advance.

Definition 1.1 (Standalone sector)**.** A sector of RG deserves its own geometry iff it satisfies all of the following.

(S1) *Identification.* It is invisible to the protocols that define the existing geometry, yet identifiable by a declared extension of them.

(S2) *Irreducibility.* Its invariants are not functions of the existing geometry.

(S3) *Intrinsic structure.* It carries structure of a different kind (connection, holonomy, curvature, topology) with invariants of its own.

(S4) *Metric lock.* Its metric is forced, or its freedom is classified and fixed by a declared RG task.

(S5) *Morphisms.* There is an admissible class of maps with an if-and-only-if monotonicity theorem.

(S6) *Collapse loci.* The conditions under which it vanishes or degenerates are characterized.

Status labels. CONSTRUCTED (proved here, battery-witnessed), IMPORTED (published theorem), DESTRUCTOR (alternatives eliminated), CONDITIONAL (holds under a stated gate). Battery groups are cited as [K*n*].

2 Setting: record cone, frame, reference quotient

Definition 2.1.

- The *record cone* V_+ is the cone of unnormalized states. Records are linear functionals on it, given by effects.
- A *frame* is a maximal family $\{e_1, \dots, e_d\}$ of mutually orthogonal minimal idempotents (in quantum language, a rank-one projective measurement). Its *populations* are $p_i = \langle e_i, \rho \rangle$.
- The *reference quotient* is $\rho \mapsto p$.
- *Dephasing* Δ maps ρ to the frame-diagonal state with the same populations.

Imported Theorem 2.2 (Symmetric cones; Jordan–von Neumann–Wigner, Koecher–Vinberg [1, 2])**.** *A homogeneous self-dual cone is the cone of squares of a Euclidean Jordan algebra. Relative to a Jordan frame, the algebra has a Peirce decomposition* $V = \bigoplus_i V_{ii} \oplus \bigoplus_{i<j} V_{ij}$ *with* $\dim V_{ii} = 1$ *and* $\dim V_{ij} = k$.

- $k = 0$*: commutative, i.e. a simplex;*
- $k = 1$*: real symmetric matrices;*
- $k = 2$*: complex Hermitian matrices;*
- $k = 4$*: quaternionic Hermitian matrices;*
- *spin factors and the exceptional algebra complete the list.*

3 S1: identification

Theorem 3.1 (Existence criterion)**.** *[K2] A maximal frame is informationally complete iff the record cone is a simplex. In the matrix cases it leaves exactly* $k\,d(d-1)/2$ *real directions unresolved, which are the Peirce off-diagonal spaces.*

Proof. The frame records are the d diagonal Peirce components. The state space has dimension $d + k\,d(d-1)/2$, so the kernel of the record map is $\bigoplus_{i<j} V_{ij}$. A simplex has $k = 0$.

Witness: an exact rank count for $k = 1, 2, 4$ and $d = 2, \dots, 5$. For example, at $d = 5$ the gaps are 10, 20 and 40. The simplex gap is 0. □

Definition 3.2. The *coherence* of ρ relative to a frame is its component in $\bigoplus_{i<j} V_{ij}$, i.e. $\rho - \Delta\rho$.

Proposition 3.3 (Collisions and complete identification)**.** *[K1] Coherence is not identified by any of the following:*

- population records: *ρ and $\Delta\rho$ share every population, at trace distance* 0.35*;*
- X,Z-only records: *they miss the imaginary quadrature (trace distance* 0.3 *at zero record gap);*
- scalar coherence values: *two states with equal $\ell_1 = 0.8$ and equal relative-entropy coherence lie at trace distance* 0.8*;*
- pair magnitudes: *two qutrits with identical populations and $|\rho_{ij}|$, and $\ell_1 = 0.48$, differ only in their loop phase, at trace distance* 0.113*.*

The d frame settings plus two quadrature settings per pair, $(|i\rangle + |j\rangle)/\sqrt{2}$ and $(|i\rangle + i|j\rangle)/\sqrt{2}$, reconstruct ρ exactly: the maximum error is 3.5×10^{-16} over 160 states with $d = 2, \ldots, 5$.

S1 holds. Coherence is invisible to the protocol that defines the population (uncertainty) geometry, and it is identifiable by a declared extension.

4 The reference quotient and the bundle

Proposition 4.1 (Dephasing is the twirl over the frame gauge group)**.** *[K3] Δ is a conditional expectation onto the frame algebra: it is idempotent, Hilbert–Schmidt self-adjoint, and a bimodule map. It equals the twirl over the frame gauge group:*

- *$\mathbb{Z}_2^d$ for real cones;*
- *$U(1)^d$ for complex cones (exact with cube roots of unity);*
- *$Sp(1)^d$ for quaternionic cones (exact with the quaternion group Q_8).*

All errors are at most 3.3×10^{-15}.

So coherence is precisely what is erased when the reference phase is forgotten: it is the fibre of the dephasing map. That fibre is *not* a single gauge orbit. The states $\left(\begin{smallmatrix} 0.5 & 0.1 \\ 0.1 & 0.5 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0.5 & 0.4 \\ 0.4 & 0.5 \end{smallmatrix}\right)$ have the same dephasing but spectra $\{0.4, 0.6\}$ and $\{0.1, 0.9\}$, so they lie in different orbits. A dephasing fibre is a union of gauge orbits, and Theorem 5.1 classifies the orbits inside it.

Proposition 4.2 (Bundle form)**.** *[K15] Δ projects the state space onto the population simplex, which is the base and carries the Fisher (uncertainty) geometry.*

(a) The fibre over p is the set of coherences with $|\rho_{ij}|^2 \le p_i p_j$. Pure states saturate the bound (to 8×10^{-17}).

(b) The fibre collapses where $p_i = 0$: there the simplex walls are stratification walls of the bundle.

(c) At incoherent states every monotone metric splits orthogonally,

$$K_f(X, X) = \sum_i \frac{X_{ii}^2}{p_i} + \sum_{i \neq j} c_f(p_i, p_j)\, |X_{ij}|^2, \qquad c_f(x, y) = \frac{1}{y\, f(x/y)},$$

into base Fisher plus fibre coefficients set by the base point (split error 6×10^{-16}).

(d) Away from the zero section, base and fibre directions mix: the cross terms reach 99%*.*

5 S2–S3: connection, holonomy and its physical weight

Under a frame gauge transformation $D = \mathrm{diag}(e^{i\theta_1}, \ldots, e^{i\theta_d})$, the entries transform as $\rho_{ij} \mapsto e^{i(\theta_i - \theta_j)}\rho_{ij}$. On the *support graph* G_ρ (edges where $\rho_{ij} \neq 0$), the normalized entries $\rho_{ij}/|\rho_{ij}|$ are therefore the parallel transports of a $U(1)$ connection.

Theorem 5.1 (Gauge classification)**.** *[K4] Two states are frame-gauge equivalent iff they have equal populations, equal magnitudes $|\rho_{ij}|$ (hence equal support graphs), and equal holonomies on a cycle basis. The holonomy of a cycle $i_1 \to \cdots \to i_m \to i_1$ is $\prod \rho_{i_a i_{a+1}}/|\rho_{i_a i_{a+1}}|$. Consequently:*

- *the number of removable phases is $|V| - c$;*
- *the number of invariant loop phases is $b_1 = |E| - |V| + c$, where c is the number of components;*
- *for the complete graph, $b_1 = (d-1)(d-2)/2$.*

Proof. Choose a spanning forest. Its tree edges fix every θ up to one constant per component. Each non-tree edge closes exactly one fundamental cycle, and its transport is then fixed iff that cycle's holonomy agrees. The gauge classes are therefore $\mathrm{Hom}(\pi_1(G_\rho), U(1)) \cong U(1)^{b_1}$.

Witness: 60 random sparse states with $d = 4, \ldots, 7$ and b_1 up to 10.

- The incidence rank equals $|V| - c$ in every case.
- The invariants are preserved to 6.7×10^{-16}.
- The gauge is reconstructed from the spanning tree to 1.1×10^{-16}.
- Changing one loop phase shifts its holonomy by at least 0.69, giving an inequivalent state (canary).

□

Proposition 5.2 (Other structure groups)**.** *[K4]*

- *For real cones the holonomy is a sign ($\mathbb{Z}_2$).*
- *For quaternionic cones it is a non-abelian $Sp(1)$ element, gauge-invariant only through its conjugacy class. Its trace is preserved to 1.8×10^{-16} while the matrix itself changes by 0.31, and holonomies of different cycles do not commute (commutator 0.24).*

Proposition 5.3 (Holonomy is physical)**.** *[K5] Take equal populations $\frac{1}{3}$ and equal pair magnitudes c with loop phase Θ. Positivity allows*

$$c \le c_{\max}(\Theta) = \Big[6 \max_k \big(-\cos \tfrac{\Theta + 2\pi k}{3} \big)\Big]^{-1} : \qquad c_{\max}(0) = \tfrac{1}{3}, \quad c_{\max}(\tfrac{\pi}{2}) = \tfrac{1}{3\sqrt{3}}, \quad c_{\max}(\pi) = \tfrac{1}{6}.$$

Bisection agrees with the formula to 10^{-12}.

S2 and S3 hold. The holonomy is invisible to populations and to every pair magnitude, so it is not a function of the uncertainty geometry. It is the invariant of a connection, and it constrains the fibre itself.

6 S4: the metric, its freedom and its selection

Imported Theorem 6.1 (Morozova–Čencov–Petz [3, 4])**.** *The Riemannian metrics on faithful states that are monotone under all channels are exactly* $K_f(X,X) = \sum_{ij} c_f(\lambda_i,\lambda_j)|X_{ij}|^2$ *in the eigenbasis of* ρ*, with* f *operator monotone,* $f(x) = xf(1/x)$ *and* $f(1) = 1$*.*

Proposition 6.2 (Čencov holds on the base, fails on the fibre)**.** *[K6]*

(a) On directions commuting with ρ*, every* K_f *equals the Fisher metric (maximum disagreement* 7.8×10^{-14}*).*

(b) On coherence directions they differ, in the order SLD $\leq$ *WY* $\leq$ *BKM* $\leq$ *max. The ordering holds in 300 of 300 cases, and the ratio max/SLD reaches* 218*.*

(c) Each is monotone under 300 random channels and 300 partial traces with extreme spectra (relative increase at most 2.4×10^{-13}*, i.e. numerical equality on isometries).*

(d) Canary: *the Hilbert–Schmidt metric grows by a factor of* 3 *under a partial trace.*

So the freedom of the quantum metric lives *entirely* in the coherence fibre. RG removes it by its own principle: geometry is what protocols resolve.

Theorem 6.3 (The resolution task selects SLD)**.** *[K7] Define the resolution metric* $g_{\mathrm{res}}(\rho, X) = \sup_M F_M(\rho, X)$*, the supremum over protocols* M *of record Fisher information. Then* $g_{\mathrm{res}} = K_{\mathrm{SLD}}$*, the smallest monotone metric. It is attained by measuring in the eigenbasis of the symmetric logarithmic derivative, and* $K_{\mathrm{SLD}} = 4\, g_{\mathrm{Bures}}$*.*

Proof. A protocol is a channel to a classical output. On classical outputs every monotone metric is the Fisher metric. By monotonicity, $F_M(\rho, X) \leq K_f(\rho, X)$ for every monotone f. The SLD-eigenbasis measurement attains K_{SLD} [5]. Hence $g_{\mathrm{res}} = K_{\mathrm{SLD}} \leq K_f$ for all f, and only the minimal metric equals the supremum.

Witness: attainment to 1.8×10^{-13}. Across 2000 random POVMs none exceeds K_{SLD} (maximum relative excess -8.6×10^{-4}). On coherence directions WY, BKM and max exceed SLD by at least a factor of 1.0022, so no protocol attains them. □

S4 holds, as a task lock. The freedom is classified (Petz) and confined to the fibre. It is fixed by the declared task: resolution (optimal-measurement estimation) gives SLD; Stein testing gives BKM; Chernoff testing gives Wigner–Yanase. This is the task lock of the main print (L38–L39), which states that descent alone selects no Petz member; what is new here is that the whole task dependence sits in the coherence fibre, since the base is Čencov-forced.

The resolution task also answers the allocation dependence of operational Fisher metrics: the supremum over protocols is allocation-free. What remains is exactly the joint obstruction of the next section.

7 S3: curvature, topology, Kähler structure

Proposition 7.1 (Curvature and topology)**.** *[K8]*

(a) On the pure qubit stratum, the Berry curvature $-2\,\mathrm{Im}\, Q_{\theta\varphi}$ *of the quantum geometric tensor* $Q_{ij} = \langle\partial_i\psi|(1 - |\psi\rangle\langle\psi|)|\partial_j\psi\rangle$ *integrates to* $2\pi C$ *with* $|C| = 1$ *(numerically* -1.0000026*; the sign is the orientation convention). This is a global invariant of the pure stratum* $\mathbb{CP}^1$*,* not *of a fibre: by (e), every fixed-population fibre is Lagrangian.*

(b) On mixed states, the mean Uhlmann curvature $\frac{1}{4}|\,\mathrm{Tr}\,\rho[L_1, L_2]|$ *is nonzero on coherent two-parameter families (minimum* 5.5×10^{-3} *in the sample) and identically zero on commuting (population) families.*

(c) Joint resolution is obstructed. *For any single-copy measurement on a qubit,* $\mathrm{Tr}(F_Q^{-1}F_C) \leq d-1=1$ *[6]; the sample maximum is* 0.99987. *A commuting family in* $d=3$ *attains* $2=d-1$ *with the reference measurement.*

(d) *The SLD bound is asymptotically jointly attainable iff* $\mathrm{Tr}\,\rho[L_i,L_j]=0$ *[7]* (IMPORTED).

(e) Toric identification. *On the pure stratum, the populations are the moment map of the frame torus. Their image is the simplex (Atiyah–Guillemin–Sternberg [8]). In coordinates* $\psi=\sum_j \sqrt{p_j}e^{i\theta_j}|j\rangle$*:*

- ω *vanishes on fixed-population fibres (maximum* 0 *over 100 points), so the fibres are Lagrangian tori;*
- ω *pairs base and fibre as* $\mathrm{Im}\,Q(\partial_{p_i},\partial_{\theta_j})=\frac{1}{2}\delta_{ij}$ *(error* 1.6×10^{-16}*).*

The curvature is therefore a base–fibre pairing (action–angle). For a qubit the fibre is a circle, on which every 2-form vanishes.

So the curvature measures exactly how far the coherence directions cannot be resolved together. This is the residual allocation dependence, and it is located: it lives where the curvature is nonzero.

Proposition 7.2 (Kähler pure stratum and dynamics)**.** *[K9] On horizontal vectors at a pure state, set* $g=\mathrm{Re}\langle\cdot,\cdot\rangle$, $\omega=\mathrm{Im}\langle\cdot,\cdot\rangle$ *and* $J=i$*. Then:*

- $g(Ju,Jv)=g(u,v)$, $\omega(u,v)=g(Ju,v)$ *and* $J^2=-1$*;*
- *the Schrödinger velocity* $-i(H-\langle H\rangle)\psi$ *is the Hamiltonian vector field of* $\langle H\rangle$*:* $2\omega(v_H,u)=d\langle H\rangle(u)$*.*

The maximum error over 200 random points is 3.8×10^{-15}*. The Kähler property of Fubini–Study is imported [9].*

S3 holds in full. The uncertainty geometry of a simplex is Riemannian. The coherence fibre adds a connection with holonomy. The bundle as a whole adds a curvature 2-form with a quantized integral and, on its pure stratum, a compatible complex and symplectic structure. That structure pairs populations with phases and generates the dynamics.

8 S5: morphisms

Theorem 8.1 (Monotonicity iff MIO)**.** *[K10] Let* $C_r(\rho)=S(\Delta\rho)-S(\rho)=\min_{\sigma\ \text{incoherent}} D(\rho\|\sigma)$*. A channel* Φ *satisfies* $C_r(\Phi\rho)\leq C_r(\rho)$ *for all* ρ *iff* Φ *maps incoherent states to incoherent states (MIO).*

Proof. ($\Leftarrow$) $C_r(\Phi\rho)\leq D(\Phi\rho\|\Phi\Delta\rho)\leq D(\rho\|\Delta\rho)$, using $\Phi\Delta\rho$ incoherent and data processing. ($\Rightarrow$) If some incoherent σ has $\Phi\sigma$ coherent, then C_r rises from 0.

Witness: 1500 checks on strictly incoherent operations, classical stochastic maps after dephasing, and mixtures of the two. The maximum increase is -0.031 and no coherence is created. *Canary:* a Haar unitary creates 1.58 bits from an incoherent state. □

S5 holds. The admissible morphisms are the channels that create no coherence from the reference section. In RG terms, these are the operations that preserve the section of the reference quotient.

9 The RG reading: coherence is the resolvability of the reference

Proposition 9.1 (Orbit resolution). *[K11] Let the reference be generated by a nondegenerate H that is diagonal in the frame. The orbit $e^{-iHt}\rho e^{iHt}$ has resolution*

$$F_H(\rho) = \sum_{k,l} \frac{2(\lambda_k - \lambda_l)^2}{\lambda_k + \lambda_l} |H_{kl}|^2.$$

- *It vanishes iff ρ is incoherent: it is 0 on 50 incoherent states and at least 0.174 on 50 coherent ones.*
- *It equals $4\,\mathrm{Var}(H)$ on pure states (to 3×10^{-9}).*
- *It does not increase under covariant channels (maximum change -0.108).*

Canary: *for a degenerate reference, coherence inside the degenerate block ($C_r \approx 1$) has orbit resolution 0.*

Reading. In RG, a relabeling is null when no record resolves it. The Tensor Derivation uses exactly this to quotient spacetime relabelings. Coherence is the opposite case: the reference relabeling *is* resolved. A state descends to the reference quotient iff its coherence vanishes. Coherence geometry is therefore the geometry of a gauge that is not null.

10 S6: collapse loci and relativity

Proposition 10.1 (Where coherence geometry vanishes or degenerates). *[K2, K11, K12, K15]*

(a) In a simplex realization ($k = 0$) the fibre is a point.

(b) Every state is incoherent in its own eigenframe ($C_r \leq 4.4 \times 10^{-16}$). A pure state reaches the maximum $\log_2 d$ in a frame unbiased to it (2.0 for $d = 4$; random frames reach at most 1.95). Coherence is a property of (state, frame) pairs, not of states.

(c) A degenerate reference sees no coherence inside its degenerate blocks.

(d) The fibre collapses at the walls $p_i = 0$ of the base simplex.

S6 holds.

11 Which structure group, and which ontology

Destructor 11.1 (Structure group, conditional on composition). [K13] With tensor-product composition, local tomography ($K_{AB} = K_A K_B$, where $K = d + k\,d(d-1)/2$) holds in every dimension for the classical simplex ($k = 0$). Among non-simplicial matrix cones it holds only for $k = 2$:

- real cones ($\mathbb{Z}_2$) have too many joint parameters, 10 versus 9;
- quaternionic cones ($Sp(1)$) have too few, 28 versus 36.

For same-type composites, $4 + 6k = (2 + k)^2$ forces $k \in \{0, 2\}$. Composition therefore does *not* supply non-simpliciality. Given non-simpliciality, the surviving structure group is $U(1)$. Network Bell tests reject real-number quantum theory with tensor-product composition [16, 17, 18].

Caveat. A real formulation with a modified composition rule has been reported to reproduce the complex predictions [19]. The lock is therefore relative to the declared composition. The operational content (the records) stays that of the complex cone.

Proposition 11.2 (Realization canary)**.** *[K14] A classical, partially coherent three-mode wave field realizes the same coherence geometry. With 200,000 samples:*

- *the coherency matrix is recovered to* 7.4×10^{-4}*;*
- *the loop holonomy is* 0.8993 *against a target of* 0.9*;*
- *analyser intensities follow* $v^\dagger J v$ *(*0.3747 *against* 0.375*).*

So coherence geometry is a property of the record cone. Quantum-specific content (entanglement, Bell violation) enters through composition, not through single-system coherence.

12 Verdict

Criterion	Result	Groups
S1 Identification	invisible to population, partial and scalar records; complete with d^2 settings; exists iff the cone is not a simplex	K1, K2
S2 Irreducibility	holonomy and magnitudes are not functions of the base; holonomy limits positivity	K1, K4, K5
S3 Intrinsic structure	connection with gauge classes $U(1)^{b_1}$; total-space Berry–Uhlmann curvature, $\lvert C\rvert = 1$; toric Kähler pure stratum with Lagrangian fibres	K4, K8, K9
S4 Metric lock	Petz freedom confined to the fibre; task lock, with the resolution task selecting SLD/Bures	K6, K7
S5 Morphisms	C_r monotone iff MIO	K10
S6 Collapse loci	simplex; eigenframe; degenerate reference; simplex walls	K2, K11, K12, K15

Theorem 12.1 (Coherence geometry)**.** *Let the record cone be non-simplicial and let a frame be declared. Then coherence is the fibre of the reference quotient (dephasing) over the population simplex. It carries:*

- *a connection under the frame gauge group, whose gauge classes are populations, magnitudes, support graph and holonomy in* $U(1)^{b_1}$ *(*$\mathbb{Z}_2$ *or conjugacy classes of* $Sp(1)$ *for the other cones);*
- *a Petz metric fixed by the declared task (SLD/Bures for resolution) which splits orthogonally from the base Fisher metric on the zero section;*
- *on the total space, a curvature 2-form that obstructs joint resolution and has quantized integrals; on the pure stratum, a toric Kähler structure whose fixed-population fibres are Lagrangian;*
- *a Kähler structure on its pure stratum, whose Hamiltonian flows are the Schrödinger dynamics.*

Its morphisms are the MIO channels. It deserves standalone status as the fibre geometry over uncertainty geometry*.*

Hierarchy. Indistinguishability (primitive) → uncertainty geometry (Fisher; Čencov-forced; universal) → coherence geometry (the fibre of the reference quotient; present exactly in non-simplicial realizations). Uncertainty is the more universal structure. Coherence is the more structured one.

Non-claim 1. No new mathematical primitive is introduced. Peirce decompositions, Petz metrics, Braunstein–Caves, Berry–Uhlmann curvature, Fubini–Study Kähler geometry, graph gauge classes and MIO monotonicity are all established. The content here is their placement as one RG sector, with the standalone test and its canaries.

Non-claim 2. Quantum theory is not derived. The non-simplicial cone, the frame and the composition rule are declared. Classical waves share the single-system geometry.

Non-claim 3. The structure group $U(1)$ is selected only together with tensor-product composition.

13 Ledger

Item	Status	Basis
Existence iff non-simplicial; Peirce gap	CONSTRUCTED	Thm. 3.1; [1, 2]
Collisions; complete pair protocols	CONSTRUCTED	Prop. 3.3
Dephasing = conditional expectation = gauge twirl	CONSTRUCTED	Prop. 4.1
Bundle over the population simplex	CONSTRUCTED	Prop. 4.2
Gauge classes; holonomy; non-abelian case	CONSTRUCTED	Thm. 5.1
Holonomy constrains positivity	CONSTRUCTED	Prop. 5.3
Petz family on the fibre	IMPORTED + witnessed	Prop. 6.2
Resolution task selects SLD (Stein: BKM; Chernoff: WY)	CONSTRUCTED + task lock	Thm. 6.3; [5]
Curvature, Chern number, joint obstruction	CONSTRUCTED + IMPORTED	Prop. 7.1; [6, 7]
Kähler pure stratum; Hamiltonian dynamics	CONSTRUCTED + IMPORTED	[9]
C_r monotone iff MIO	CONSTRUCTED	Thm. 8.1
Coherence = orbit resolution	CONSTRUCTED	Prop. 9.1
Collapse loci; frame relativity	CONSTRUCTED	Prop. 10.1
Structure group $U(1)$	DESTRUCTOR, CONDITIONAL on composition	Destr. 11.1; [16, 17, 18, 19]
Non-simplicial cone from RG records	OPEN	—
Composition rule from RG	OPEN	—
Infinite-dimensional and continuous-variable fibres	NOT ADDRESSED	—

14 Corrections after external audit (this edition)

- *Fibre versus orbit.* An earlier wording called dephasing "a quotient by the frame gauge group". Dephasing is the gauge *twirl*; its fibres are unions of orbits (Section 4).

- *Chern number.* An earlier wording assigned $|C| = 1$ to the fibre. It belongs to the total pure stratum; the fibres are Lagrangian (Proposition 7.1(e)).
- *Composition.* An earlier wording said local tomography selects $k = 2$ "only". The classical simplex also passes, so non-simpliciality needs a separate premise (Destructor 11.1).

15 Reproduction

- `rg_coherence_geometry.py` runs K1–K15 in a few seconds, using numpy and scipy. The seed is 20261110 and reruns are byte-identical.
- Every group carries a canary: an alternative that must fail, or a freedom that must survive.
- *Limits.*
  - Finite dimension.
  - Petz, Braunstein–Caves, Gill–Massar, Ragy et al., and the Kähler property of Fubini–Study are imported.
  - K14 is synthetic.
  - The data in Destructor 11.1 are quoted, not fitted.

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